

L03 Reflective Journal

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A.I. at the Edge & IIOT Env-17181

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Reflective Journal (L03)

1. Observations

During this exercise, I successfully changed an AI model from a high-level development environment (Python in VS Code) to a simulated hardware environment (Edge Impulse). The initial model was a CNN trained on the MNIST dataset that was then turned into a TensorFlow Lite (.tflite) format. The results of simulation on a Cortex-M4F 80MHz processor offered critical insight into the physical footprint of the model:

- **Processing Time:** 7 ms
- **Peak RAM Usage:** 4 KB
- **Flash Usage:** About 2.7 MB

2. Technical Challenges and Solutions

The most significant challenge I encountered was in Step 5: Test and Validate phase in the Edge Impulse platform. I faced a "Job failed" error claiming that the labeling method was set to "Bounding boxes (Object detection)" as I tried to run a standard model of classification. I resolved this by manually updating the Data Acquisition tab to make sure my samples were labeled correctly as "digit" and that the dashboard of the project was in perfect configuration for classification instead of object detection. This was an essential step for unlocking the "On-device performance" metrics required for the assignment.

3. Personal Reflections

In this exercise, I learned that "Edge AI" requires a different mindset than the development of traditional software. In VS Code, I focused mainly on the accuracy of training, but in Edge Impulse, I was able to prioritize Latency as well as RAM usage. I experienced the model process a digit in only 7 ms on a simulated microcontroller. This was helpful in making me comprehend how optimized models can facilitate real-time applications in the industry. I now understand that the pipeline of deployment pipeline is vital just like the code utilized in training the model itself.